

10th International Conference of Information and Communication Technology (ICICT-2020)

# Research on bandwidth allocation strategy of train communication network based on time division multiplexing

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## Abstract

In order to solve the media contention problem of train switched Ethernet and improve the real-time performance of train communication network, this paper conducts real-time research on the access mode of terminal equipment of train communication network. Through the time division multiplexing mechanism, the network bandwidth is allocated according to the data type, and a bandwidth allocation strategy based on temporal constraints is established. It mainly schedules the time slots of real-time periodic data, and obtains a static schedule of multiple tasks based on the temporal constraints, so that the terminal equipment can receive the static schedule in the initial stage of the system. Each terminal device sends data in the specified time slot according to the static schedule to avoid the collision of network data frames, and at the same time reserve more bandwidth for the transmission of real-time aperiodic data, ensuring the real-time and reliability of train communication network.

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Peer-review under responsibility of the scientific committee of the 10th International Conference of Information and Communication Technology.

**Keywords:** Train communication network; switched Ethernet; time division multiplexing; bandwidth allocation; communication scheduling

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## 1. Introduction

Because the traditional shared Ethernet uses the hub as the transmission center of the train network, the device nodes on the respective hub share the same network resources while also sharing the same collision domain<sup>1</sup>. Traditional Ethernet uses the CSMA/CD media access control method at the MAC layer<sup>2</sup>. This mode is likely to cause some node data to lose packets or affect the transmission delay when the network transmission conflicts, which not meet the requirements of high real-time and high reliability of train communication network.

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Therefore, most scholars have studied the real-time performance of train communication network from multiple perspectives. Literature [3,4] adds a real-time dispatching layer on top of the MAC layer, but this method needs to modify the protocol stack; literature [5] modifies the MAC layer protocol, resulting in poor compatibility; literature [6] uses a traffic shaping method, which does not significantly reduce the delay of real-time data. There are also some scholars on the collision and conflict of network data transmission. Literature [7,8] improves the backoff algorithm of the MAC layer to reduce the waiting delay of data retransmission after collision, but this method cannot avoid the data collision occurred from the root cause. Time division multiplexing is an access control method that divides the time of each node's information transmission according to the time slot. It divides the time axis into time frames of certain length and cycle. Time division multiplexing eliminates collision and provides quality assurance, delay guarantee and differential service<sup>9</sup>. In addition, in the time division multiplexing network, each node can only send data in its allocated time slot, each node occupies one or more time slots, the same node can transmit different data in different periods. Moreover, the network does not need to control information to avoid data collision between nodes, it is a simple and mature channel allocation<sup>10</sup>, and the mechanism solves the conflict and data collision of traditional Ethernet.

In order to reduce the number of data collisions in train communication network, and improve the network real-time performance. In this paper, switched Ethernet is used to reduce the queue waiting time after transmission conflict collision instead of traditional Ethernet, and a bandwidth allocation strategy based on time division multiplexing is proposed. According to the data type, the transmission period is divided into three time slots, and the transmission problem of real-time periodic data is analyzed. The communication scheduling algorithm with temporal constraint is used to establish the real-time periodic time slot static scheduling of multiple tasks, and get the static scheduling table. Each node can send data according to the specified time in the static scheduling table, reduce the possibility of collision during information transmission, and improve the real-time and reliability of train communication network.

## 2. Problem analysis

The train communication network data includes real-time periodic data, real-time aperiodic data and non real-time data. The bandwidth allocation of train communication network mainly has the following problems: 1) increasing the transmission cycle bandwidth will lead to longer channel idle time and waste channel transmission resources; 2) there is a sudden situation in the real-time aperiodic data transmission<sup>11</sup>, so it is necessary to reserve appropriate time slots to avoid changing the original transmission cycle. On the basis of this, the real-time periodic time slot is optimized and increased; 3) when the network data transmission, some links may need to transmit multiple real-time periodic data at the same time. In order to solve the above problems, this paper allocates bandwidth according to three data types. A time division multiplexing cycle is divided into three stages, as shown in Figure 1: real-time periodic data transmission phase, real-time aperiodic data transmission phase and non real-time periodic data transmission phase.

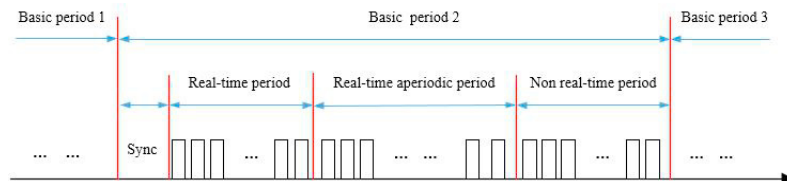


Fig. 1. Time division multiplexing cycle

Considering that the real-time periodic data is continuous and stable<sup>12</sup>, time division multiplexing mechanism is mainly used to divide the time slots of the real-time periodic data, so that makes sure the information transmission can be completed in the shortest time, which can meet the requirements of the upper limit of the delay of the real-time data transmission, and can meet the fairness of the real-time periodic data and real-time aperiodic data transmission, and reserve more time for the real-time aperiodic data transmission<sup>13</sup>. Therefore, this paper focuses on

the bandwidth allocation of real-time periodic data as the main research content, through the scheduling of real-time periodic bandwidth to ensure the upper limit of real-time data delay.

### 3. Communication model of real-time periodic data

In order to ensure that the real-time data transmission has the upper limit of delay, this paper introduces four vehicles and uses the topology structure of ring connection to allocate the bandwidth of real-time periodic data, and establishes the switched Ethernet train network topology diagram as shown in figure 2(a). The four consist switches (CS) in figure 2(a) represent the four vehicles respectively, and the consist switch 1 (CS1) is connected with the central control unit (CCU) of the main node, other consist switches are connected to the slave node electronic control unit (ECU). In order to further explain the data communication process, figure 2(a) is refined into figure 2(b) train network point-to-point communication model. In figure 2(b), node 1 (N1) is the central control unit (CCU), node 2 (N2), node 3 (N3) and node 4 (N4) are each electronic control unit (ECU), the communication between each node is completed by the consist switch. The consist switches to which nodes N1, N2, N3 and N4 belong are CS1, CS2, CS3 and CS4 respectively, each switch has four ports which are P1, P2, P3 and P4.

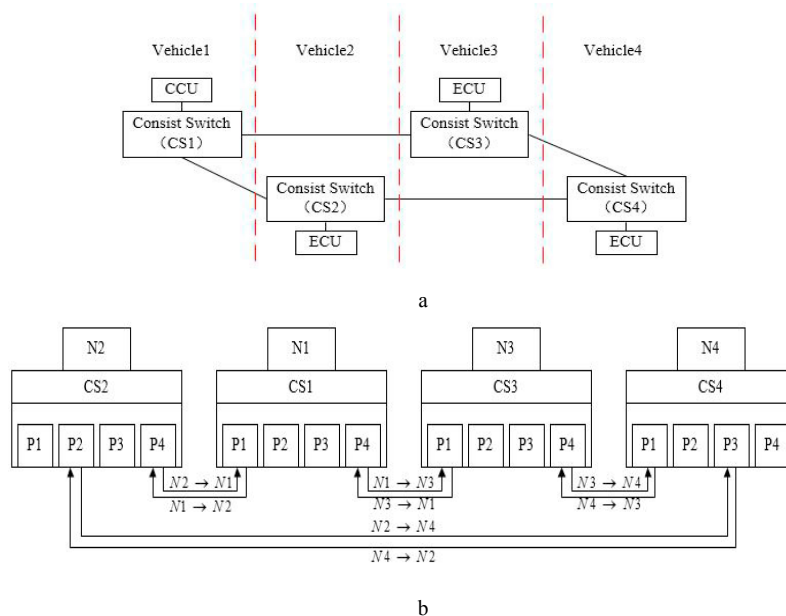


Fig. 2. (a) Train network topology of Switched Ethernet; (b) Point-to-point communication model of train communication network

Suppose that there are five messages M1-M5 that need to be transmitted in the network in figure 2(b). The message transmission and task allocation are shown in table 1. It can be seen from figure 2(b) that there is no direct link between the source node and the destination node of M2 and M3 for data transmission, so the transmission of M2 and M3 needs to be divided into two tasks, that is, the transmission of a message needs to be completed by multiple tasks. In addition, it can be found from table 1 that multiple tasks preempt the same link to transmit data. In this case, it is necessary to establish a reasonable scheduling method to ensure that only one task can use a certain link at the same time.

Table 1. Information and task list of train communication network

| Message | Source node | Destination node | Task composition                     |
|---------|-------------|------------------|--------------------------------------|
| M1      | N1          | N2               | T1: N1→N2                            |
| M2      | N1          | N4               | T2: N1→N2/ N1→N3<br>T3: N2→N4/ N3→N4 |
| M3      | N4          | N1               | T4: N4→N2/ N4→N3<br>T5: N2→N1/ N3→N1 |
| M4      | N2          | N1               | T6: N2→N1                            |
| M5      | N3          | N1               | T7: N3→N1                            |

From the analysis, the main purpose of solving the problem is to get the maximum time slot upper limit of the real-time period in the bandwidth of time division multiplexing by scheduling algorithm, and reserve more time slots for real-time aperiodic data. In short, the main research content of this paper is how to complete the transmission of five messages in the shortest time, and ensure that each message starts to execute the task within its time period and completes the data transmission within the deadline.

#### 4. Bandwidth allocation strategy based on temporal constraints

All In this paper, a bandwidth allocation strategy based on temporal constraints is proposed to schedule the real-time periodic data in the transmission process. Temporal constraint refers to the three time factors of a message in the transmission process, namely the message release time  $r$  (the earliest time the first bit of the message is sent by the source node), the message deadline  $d$  (the time when the last bit of the message arrives at the destination node and is received), and the maximum allowable execution time of message transmission  $D$ .

##### 4.1. Network delay parameters

Before the communication scheduling analysis, it is necessary to explain the network delay parameters. The main network delay is divided into node processing delay and link transmission delay.

###### (1) Nodal processing delay $P_{Ni}$

The node processing delay  $P_{Ni}$  mainly includes the node sending delay  $T_{ND}$  and the minimum time interval between tasks (this paper is stipulated by IEEE 802.3 standard, and considering the time margin, the minimum time interval between tasks is 960ns). Among, the node transmission delay is

$$T_{ND} = L_{bit} \times Bit\_time \quad (1)$$

In the formula,  $L_{bit}$  is the length of the sent data frame, that is, the number of bits in the data frame, and  $Bit\_time$  is the time required to send a unit bit. In this paper, 100Mbit/s Ethernet is used, so the value  $Bit\_time$  is 10ns.

###### (2) Link transmission delay $T_{LD}$

$$T_{LD} = T_{TxD/Ni} + T_{RxD/Nj} + T_{CD} + T_{BD/Nj} + T_{\gamma} \quad (2)$$

In the formula,  $T_{TxD/Ni}$  is the processing delay of the send port of the switch on the node  $N_i$  side,  $T_{RxD/Nj}$  is the processing time of the switch receiving port on the node  $N_j$  side,  $T_{BD}$  is the basic delay of the switch bridge on the node  $N_j$  side, and  $T_{CD}$  is the cable delay (take the length of the carriage as 20m, the transmission rate of the cable is 5ns/m),  $T_{\gamma}$  is the margin value, generally taken as 1000ns.

#### 4.2. Communication scheduling analysis

All tasks in the network message transmission are regarded as a task sequence set  $T = \{T_1, T_2, \dots, T_n\}$ , and the start time of each task is recorded as the task execution time sequence set  $S = \{S_{T1}, S_{T2}, \dots, S_{Tn}\}$ . Under the temporal constraints of three factors, the communication scheduling content is divided into five situations according to the task transmission sequence:

##### (1) Task successive scheduling

Task  $T_j$  begins to transmit data immediately when task  $T_i$  ends, at this time, the start execution time of task  $T_j$  is

$$\begin{cases} S_{Tj} = r_i + P_{Ni} & T_{LD/Ti} < P_{Ni} \\ S_{Tj} = r_i + T_{LD/Ti} & T_{LD/Ti} > P_{Ni} \end{cases} \quad (3)$$

Then, the last task to complete the transmission is  $T_j$ , and its deadline is

$$\begin{cases} d = S_{Tj} + P_{Nj} & T_{LD/Tj} < P_{Nj} \\ d = S_{Tj} + T_{LD/Tj} & T_{LD/Tj} > P_{Nj} \end{cases} \quad (4)$$

##### (2) Task conflict insertion scheduling

The transmission of the M1 needs to be completed through two tasks  $T_i$  and  $T_{i+1}$ . Before the transmission of the first task  $T_i$  is completed, other nodes need to occupy the transmission link of the second task  $T_{i+1}$  of the M1 to send the M2. If the M2 does not preempt the link and the M2 cannot complete the task transmission within the maximum deadline of the task, then the task  $T_j$  of the M2 is allowed to perform data transmission first.

At this point, the last completed transmission task is  $T_{i+1}$ , and its deadline is

$$\begin{cases} d = r_j + P_{Nj} + T_{LD/Ti} & T_{LD/Tj} < P_{Nj}, T_{LD/Ti} > P_{Ni} \\ d = r_j + P_{Nj} + P_{Ni} & T_{LD/Tj} < P_{Nj}, T_{LD/Ti} < P_{Ni} \\ d = r_j + T_{LD/Tj} + T_{LD/Ti} & T_{LD/Tj} > P_{Nj}, T_{LD/Ti} > P_{Ni} \\ d = r_j + T_{LD/Tj} + P_{Ni} & T_{LD/Tj} > P_{Nj}, T_{LD/Ti} < P_{Ni} \end{cases} \quad (5)$$

##### (3) Task conflict backoff scheduling

The transmission of the M1 needs to be completed through two tasks  $T_i$  and  $T_{i+1}$ . Before the transmission of the first task  $T_i$  is completed, other nodes need to occupy the transmission link of the second task  $T_{i+1}$  of the M1 to send the M2. If the M2 does not preempt the link, and the M2 can complete the task transmission within the maximum deadline of the task, then the task  $T_j$  of the M2 will first perform the node data processing, and the data will be transmitted after the second task  $T_{i+1}$  of the M1 is completed.

At this point, the last completed transmission task is  $T_j$ , and its deadline is

$$\begin{cases} d = r_i + T_{LD/Ti} + P_{Ni} + P_{Nj} & T_{LD/Tj} < P_{Nj} \\ d = r_i + T_{LD/Ti} + P_{Ni} + T_{LD/Tj} & T_{LD/Tj} > P_{Nj} \end{cases} \quad (6)$$

##### (4) Task interval scheduling

Task  $T_j$  is the last task on a certain link in all messages, and there is a certain interval with other tasks. At this time, the deadline of task  $T_j$  is

$$\begin{cases} d = r_j + P_{Nj} & T_{LD/T_j} < P_{Nj} \\ d = r_j + T_{LD/T_j} & T_{LD/T_j} > P_{Nj} \end{cases} \quad (7)$$

##### (5) Task overlap scheduling

The transmission of M1 needs to be completed through two tasks  $T_i$  and  $T_{i+1}$ , and both tasks do not conflict with other tasks. Before the transmission of its first task  $T_i$  is complete, the second task  $T_{i+1}$  begins to forward some of the received information. At this time, the task deadline is

$$d = r_i + T_{LD/T_i} + P_{Ni} \quad (8)$$

## 5. Example calculation

The switch parameters in rail transit are used for example calculation. The network delay parameters of node N1 are respectively:  $T_{TxD/N1}=1192ns$ ,  $T_{RxD/N1}=363ns$ ,  $T_{BD/N1}=1720ns$ ; the network delay parameters of node N2 are respectively:  $T_{TxD/N2}=1580ns$ ,  $T_{RxD/N2}=350ns$ ,  $T_{BD/N2}=2720ns$ ; the network delay parameters of node N3 are respectively:  $T_{TxD/N3}=1212ns$ ,  $T_{RxD/N3}=418ns$ ,  $T_{BD/N3}=1920ns$ ; the network delay parameters of node N4 are the same as those of node N3<sup>13</sup>.

According to IEEE802.3 standard, the standard Ethernet data frame length is between 64byte and 1518byte. If it is larger or smaller than these values, the switch will judge it as bad frame and discard it. Assuming that the data frame length of message M2 and M3 is 64BYTE, and that of message M1, M4 and M5 is 128byte. The transmission delay of M1-M5 is 10240ns, 5120ns, 5120ns, 10240ns and 10240ns respectively. From this, can get the task information table of each message, as shown in table 2.

Table 2. Details of the task and information

| Message | Source/destination node | Task composition | Nodal processing delay(ns) | Link transmission delay(ns) | $r$ (ns) |
|---------|-------------------------|------------------|----------------------------|-----------------------------|----------|
| M1      | N1/N2                   | T1: N1→N2        | 10240                      | 5362                        | 5000     |
| M2      | N1/N4                   | T2: N1→N2/ N1→N3 | 5120                       | 5362/4630                   | 5000     |
|         |                         | T3: N2→N4/ N3→N4 |                            | 5018/4650                   |          |
| M3      | N4/N1                   | T4: N4→N2/ N4→N3 | 5120                       | 5382/4650                   | 12000    |
|         |                         | T5: N2→N1/ N3→N1 |                            | 4763/4395                   |          |
| M4      | N2/N1                   | T6: N2→N1        | 10240                      | 4763                        | 15000    |
| M5      | N3/N1                   | T7: N3→N1        | 10240                      | 4395                        | 17000    |

By combining the data in Table 2 with the communication scheduling model in section 4.2, the static scheduling table of multi task can be obtained. The static scheduling table is represented by Gantt chart, as shown in figure 3 (the gray part is the minimum time interval between tasks).

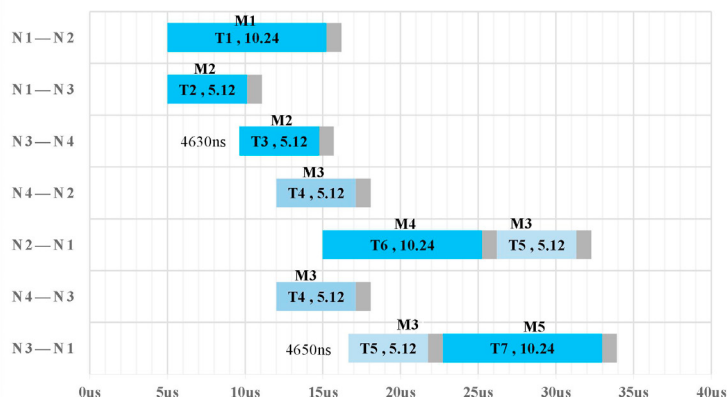


Fig. 3. Multi-task communication scheduling Gantt chart based on temporal constraints

Among, 1) M1 and M2 may have task conflict during transmission, but M2 has two path task modes, which can avoid conflict with M1. In addition, the transmission of M2 belongs to task overlap scheduling; 2) M1 occupies a single link transmission and there is no other task transmission in the link, which belongs to task interval scheduling; 3) There are two path task modes for M3 transmission, but they both conflict with M4 and M5 respectively. In this case, the maximum deadline of each message should be considered. Assuming that the deadline of M4 is 15000ns, the deadline of M5 is 17000ns, the task transmission in M3 and M4 belongs to task conflict insertion scheduling, and task transmission in M3 and M5 belongs to task conflict backoff scheduling.

It can be known from the Gantt chart that the minimum real-time cycle time slot occupied by all tasks in the five messages analyzed above is 3228 ns, and the maximum real-time cycle time slot is 3393 ns.

## 6. Conclusion

In this paper, time division multiplexing mechanism is introduced into the train Ethernet, and the bandwidth allocation strategy based on temporal constraints is established to schedule the bandwidth of real-time periodic data, and the static scheduling table of multiple tasks is obtained. It ensures that each equipment node transmits data within its own time period, solves the data collision problem in the data transmission process of traditional Ethernet, ensures the original compatibility of the network. It reserves more bandwidth for real-time aperiodic data, and also improves the real-time and reliability of train communication network.

## Acknowledgement

Fund project: Scientific Research Project of Liaoning Provincial Department of Education (JDL2019011)

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